

The Translation Between Responsible AI Research and AI Legislation

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Background

- The corpus of Responsible AI (RAI) research has **grown substantially in recent years**, yet evidence suggests its **slow translational uptake** in industry and governance (Septiandri et al.).
- Legislators globally have pursued **divergent regulatory strategies** for AI governance, with the relationship between RAI scholarship and legislative frameworks **largely underexplored**.
- Examining the translations from academic research to policy represents a critical subject for understanding the **foundation of RAI principles**.

Research Questions

- Which thematic categories of RAI research demonstrate **superior citation impact**?
- To what degree do AI regulatory frameworks **reflect core RAI principles**?
- Across the policy dataset, what is the **extent of alignment** between regulation and RAI research literature?

Methods

- Literature review of two flagship AI conferences: AI in Ethics and Society (**AIES**) and Neural Information Processing Systems (**NeurIPS**).
- RAI thematic classification of research abstracts via **semantic similarity scoring** against a keyword taxonomy from NIST (Wang et al.).
- Construction of a policy document collection spanning **major national and international AI regulations** with computed RAI thematic profiles.
- Policy-to-research alignment** quantified through matrix multiplication of keyword similarity representations (Koundouri et al.).

Data

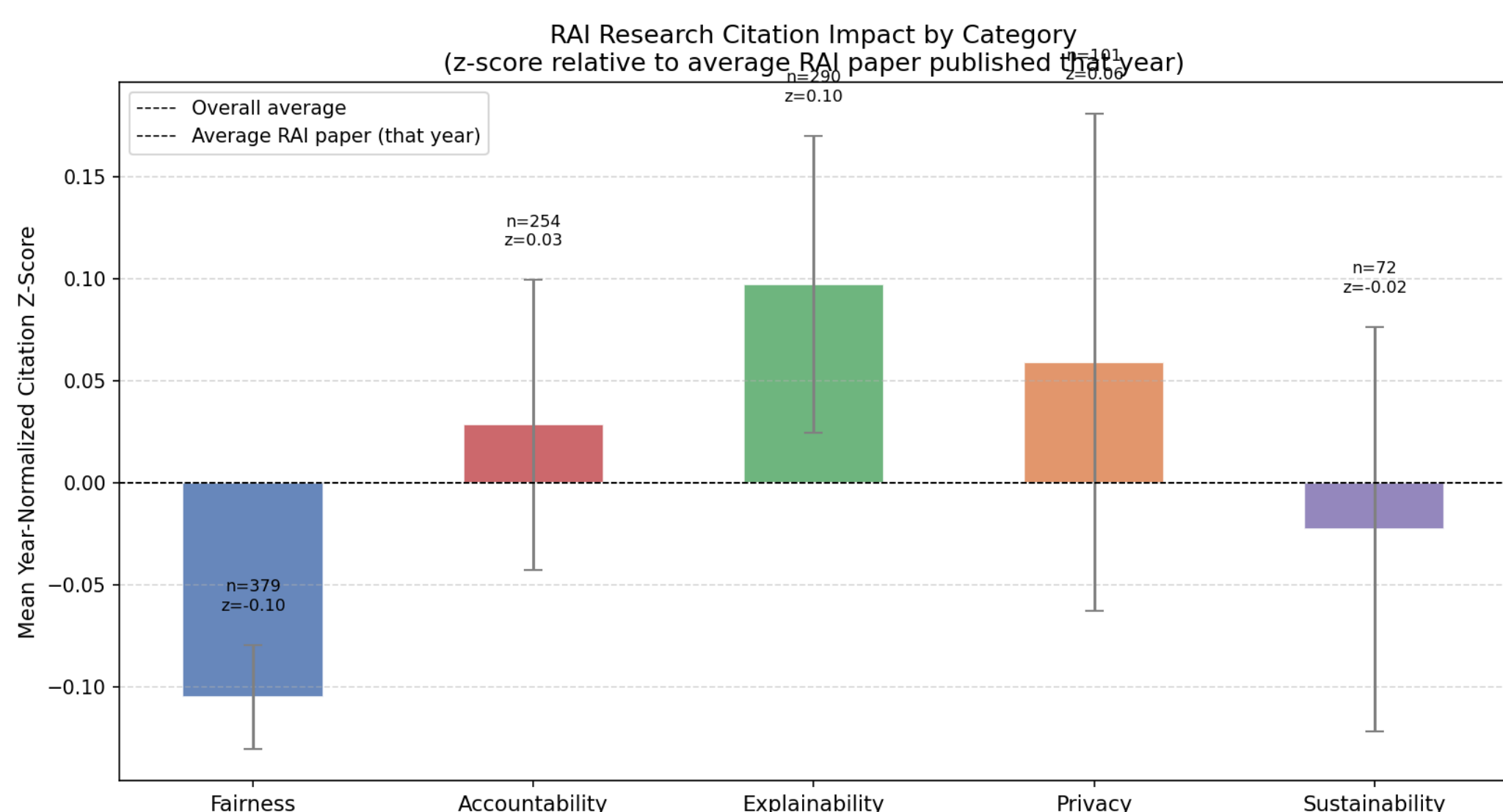


Figure 1: Mean year-normalized citation z-score by RAI category relative to RAI publications.

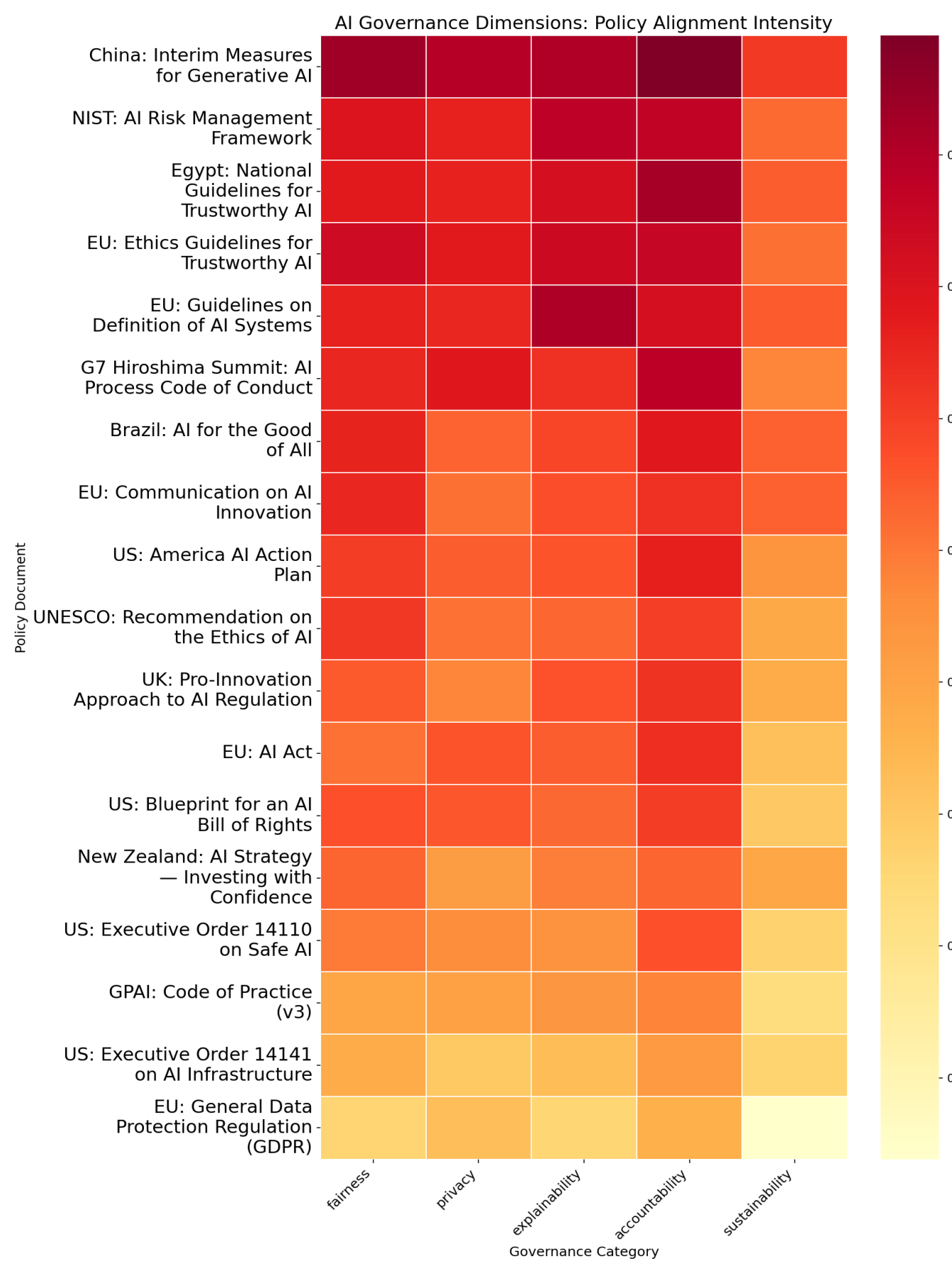


Figure 2: Thematic emphasis of AI policy documents across RAI categories derived from chunk-level similarity scoring.

Topic	Keywords
Fairness	Fairness, equality, etc.
Privacy	Confidentiality, privacy, etc.
Explainability	Explainable, etc.
Accountability	Accountable, transparent, governance, audit, etc.
Sustainability	Green, eco-friendly, etc.

Figure 3: Example keywords.

Results

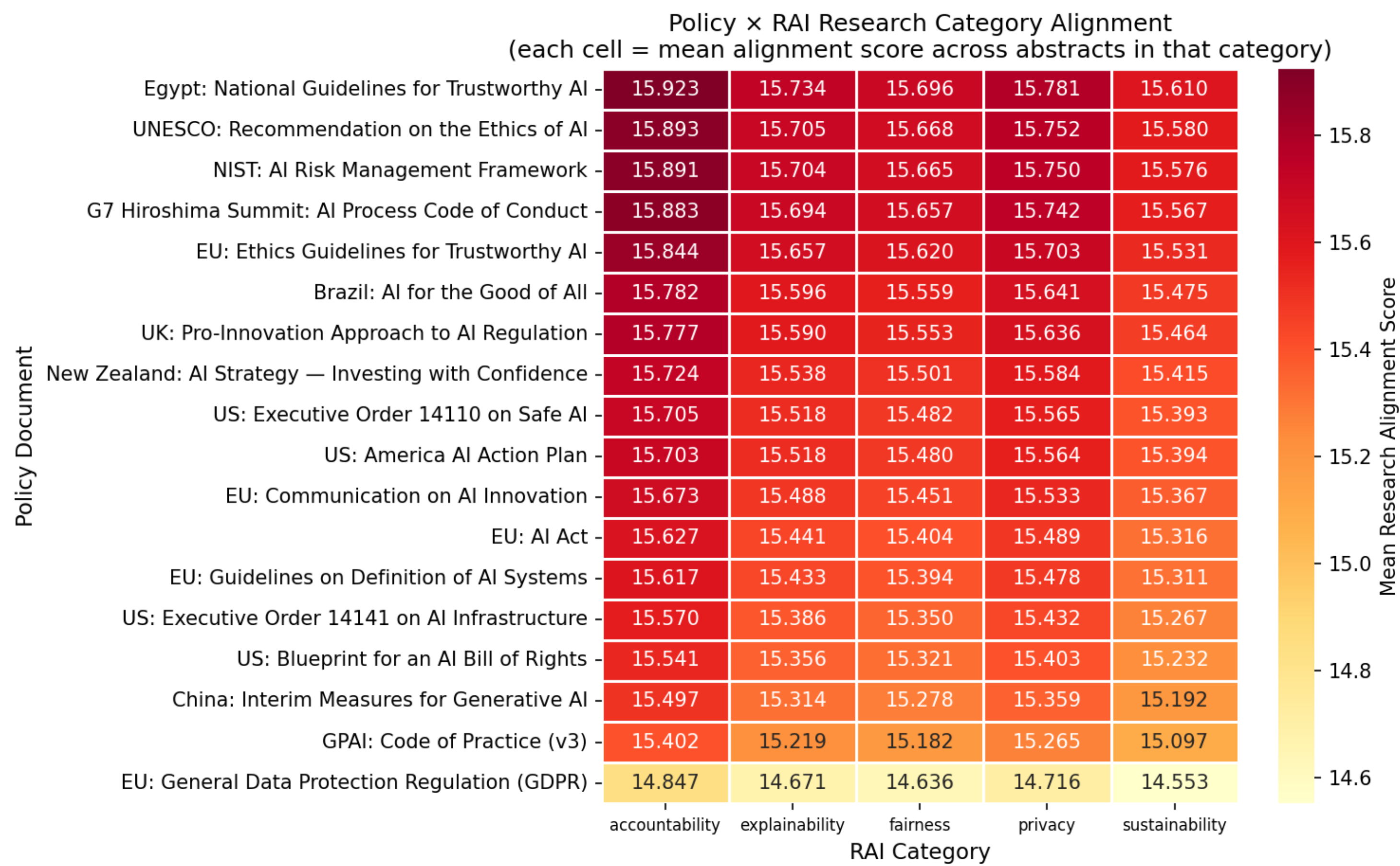


Figure 4: Mean research alignment score per policy document, by RAI category

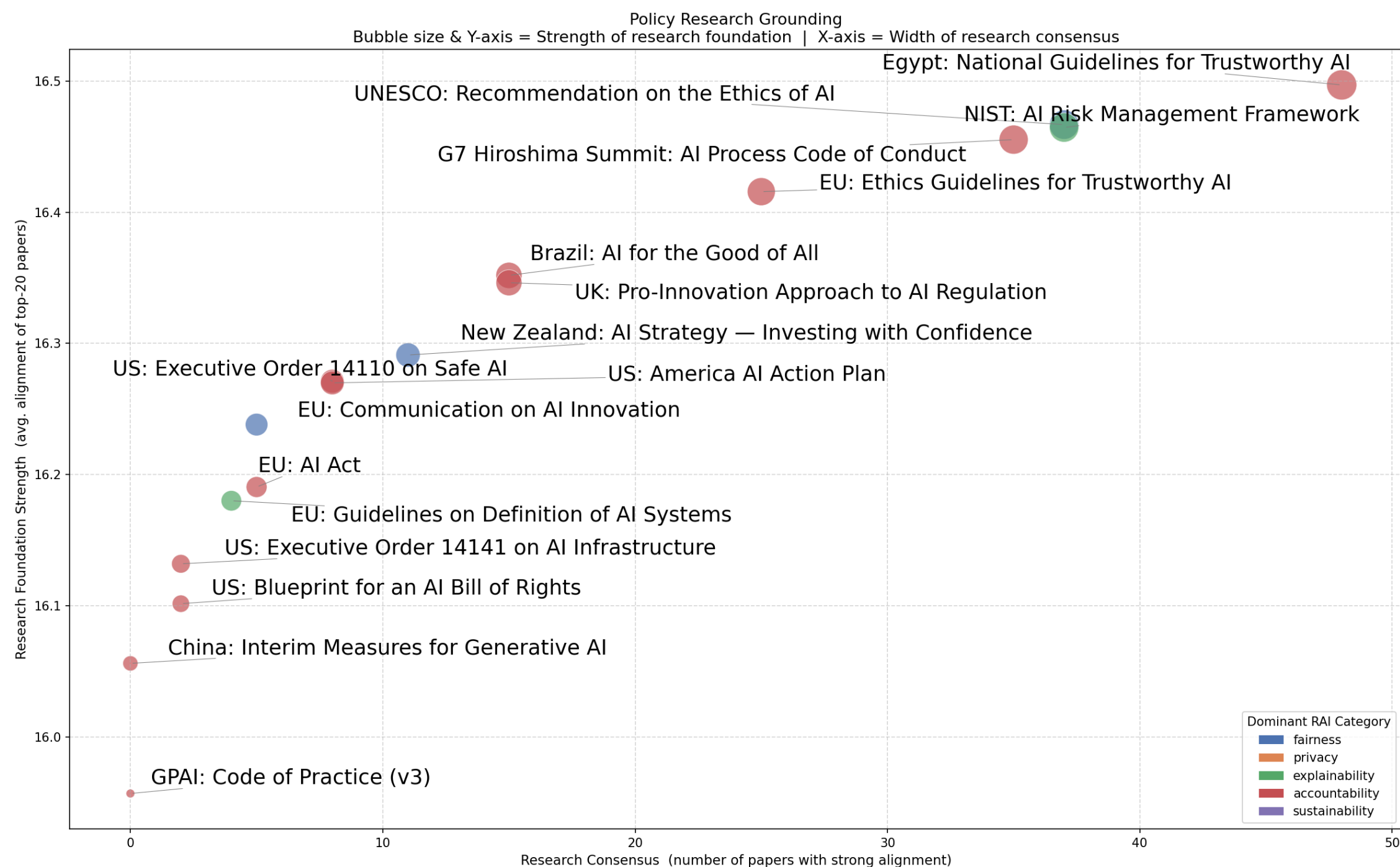


Figure 5: Policy 'grounding,' measured by strength and depth of supporting research.



Scan to access the complete policy document rankings and their highest-aligned research abstracts.

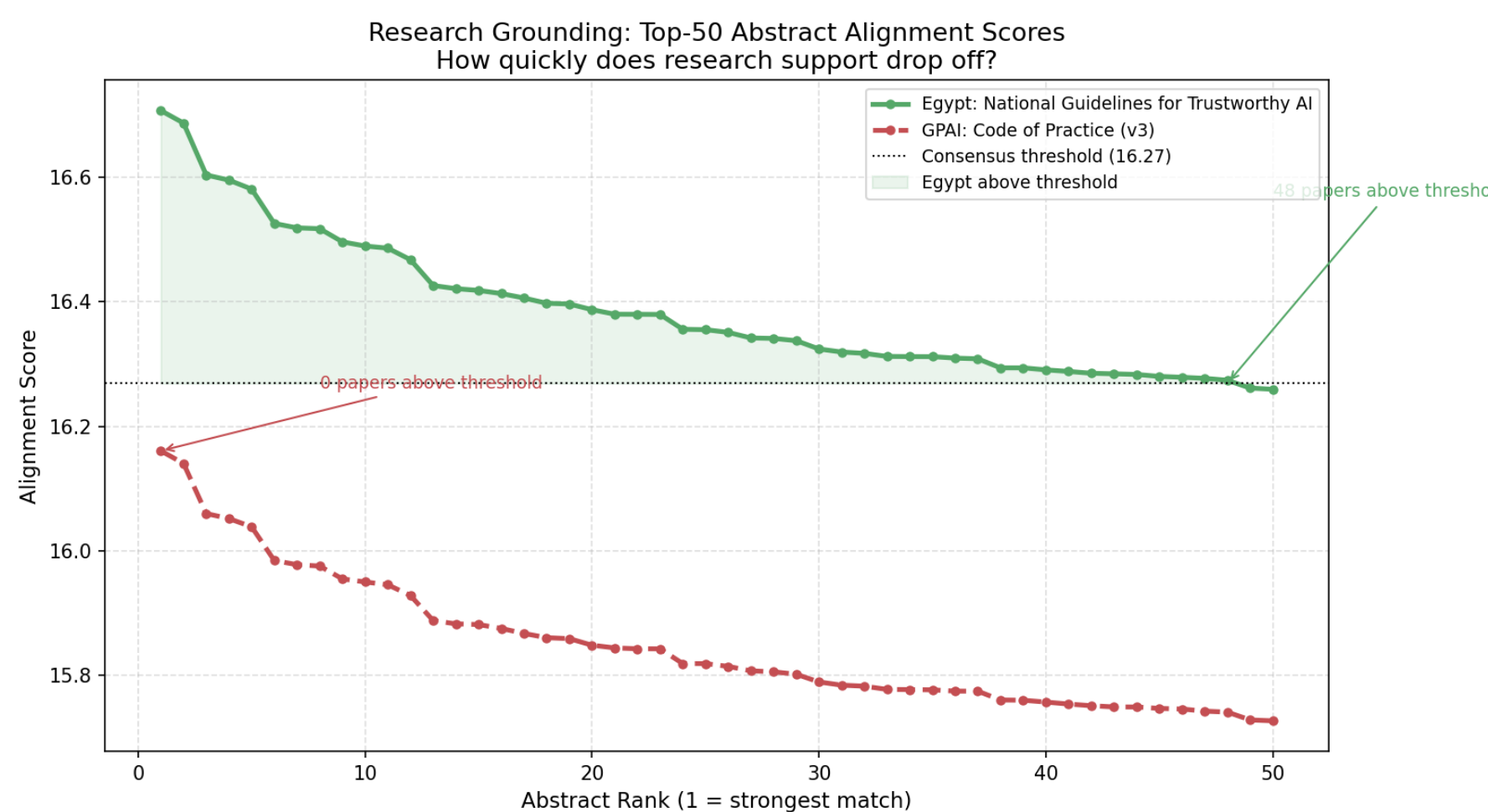


Figure 6: Ranked abstract alignment scores for high and low performing policies to illustrate the consensus gap.

What is a Semantic Similarity Score?

UNESCO: Recommendation on the Ethics of AI

Keyword matched: **Fairness (score: .883)**

"to facilitate fair access to AI technologies and address the challenges that they bring to diversity and the interconnectivity of"

Keyword matched: **Compliance (score: .750)**

"Even when importing a model from a third-party, we may still need to apply some actions"

Discussion

- Egypt** (48 papers, strongest grounding) and **UNESCO** lead in research foundation as comprehensive ethics frameworks.
- China** scores highest on RAI keyword density but has **zero papers above the research consensus threshold** which is the largest gap between topical ambition and research grounding in the dataset.
- US** policies cluster in the middle with broad enough to appear grounded but room for improvement.
- Fairness** dominates RAI paper production, **explainability** dominates paper citations, but **accountability** dominates policy language.
- Sustainability** is near absent in both research production and citation impact, despite appearances in policy documents.

Future Research

- Overton Index** usage to identify if high semantic grounding aligns with citations
- Investigate why **sustainability is a neglected realm of RAI**
- Examine the role of **standards bodies** for translational purposes.

Citations

Phoebe Koundouri & Kostas Dellis & Monika Mavragani & Angelos Plataniotis & Georgios Feretzakis, 2025. "Do SDGs Support Human Security? A Machine Learning Analysis with Policy Recommendations," DEOS Working Papers 2538, Athens University of Economics and Business. <https://ideas.repec.org/p/aeu/wpaper/2538.html>

Ali Akbar Septiandri & Marios Constantinides & Daniele Quercia (2024). "The Impact of Responsible AI Research on Innovation and Development." <https://arxiv.org/abs/2407.15647>

Liang Wang & Nan Yang & Xiaolong Huang & Binxing Jiao & Linjun Yang & Daxin Jiang & Rangan Majumder & Furu Wei (2022). "Text Embeddings by Weakly-Supervised Contrastive Pre-training." <https://arxiv.org/abs/2212.03533>